Smart V2X Cooperative Perception In Autonomous Vehicles

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01 Abstract

Cooperative perception enables autonomous vehicles to extend their awareness beyond onboard sensor limitations through shared perception data. This project proposes a lightweight framework where an Ego vehicle fuses its own detections with those from a Road Side Unit (RSU) to improve visibility under occlusion. Instead of transmitting raw images, only detection results are shared to reduce bandwidth usage. A confidence-triggered activation strategy further minimises unnecessary communication.

02 Problem Statement

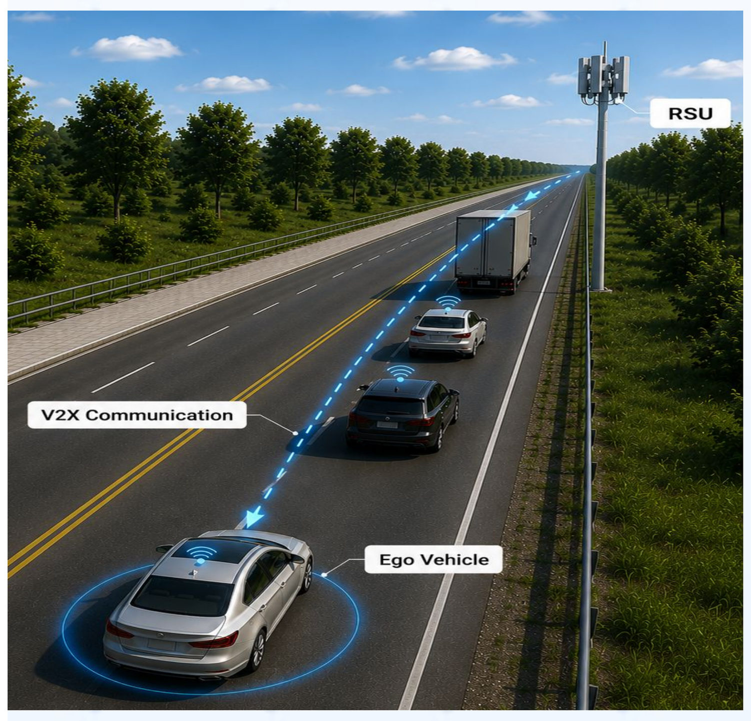

Figure 1: Cooperative V2X communication between ego vehicle and roadside unit (RSU)

Autonomous vehicles face blind spots caused by trucks, buildings and road bends, limiting hazard detection. Existing V2X cooperative perception approaches often require continuous high-volume sensor transmission, causing high communication overhead. This project proposes a confidence-triggered strategy to reduce unnecessary communication while recovering occluded detections.

03 Project Objectives

R1: Baseline Detection
Establish a YOLOv8n single-camera baseline.

R2: Cooperative Perception
Enable Ego detection of occluded objects using RSU assistance.

R3: Confidence-Triggered Activation
Trigger RSU communication below a confidence threshold.

R4: Transmission Optimisation
Reduce communication overhead via efficient RSU transmission.

04 Methodology

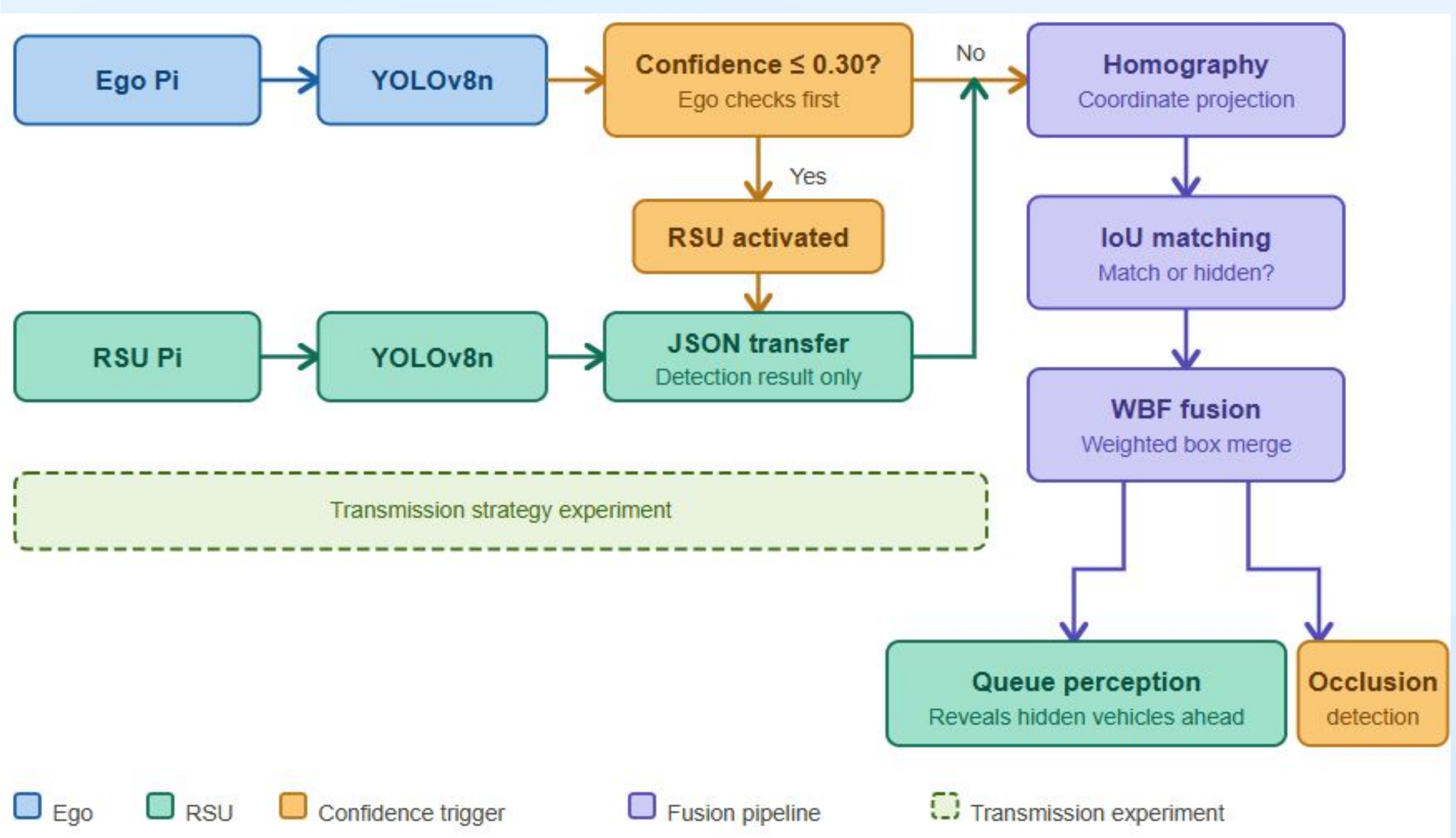

Figure 2: System architecture diagram

05 Key Highlights

Confidence-Triggered Occlusion Recovery

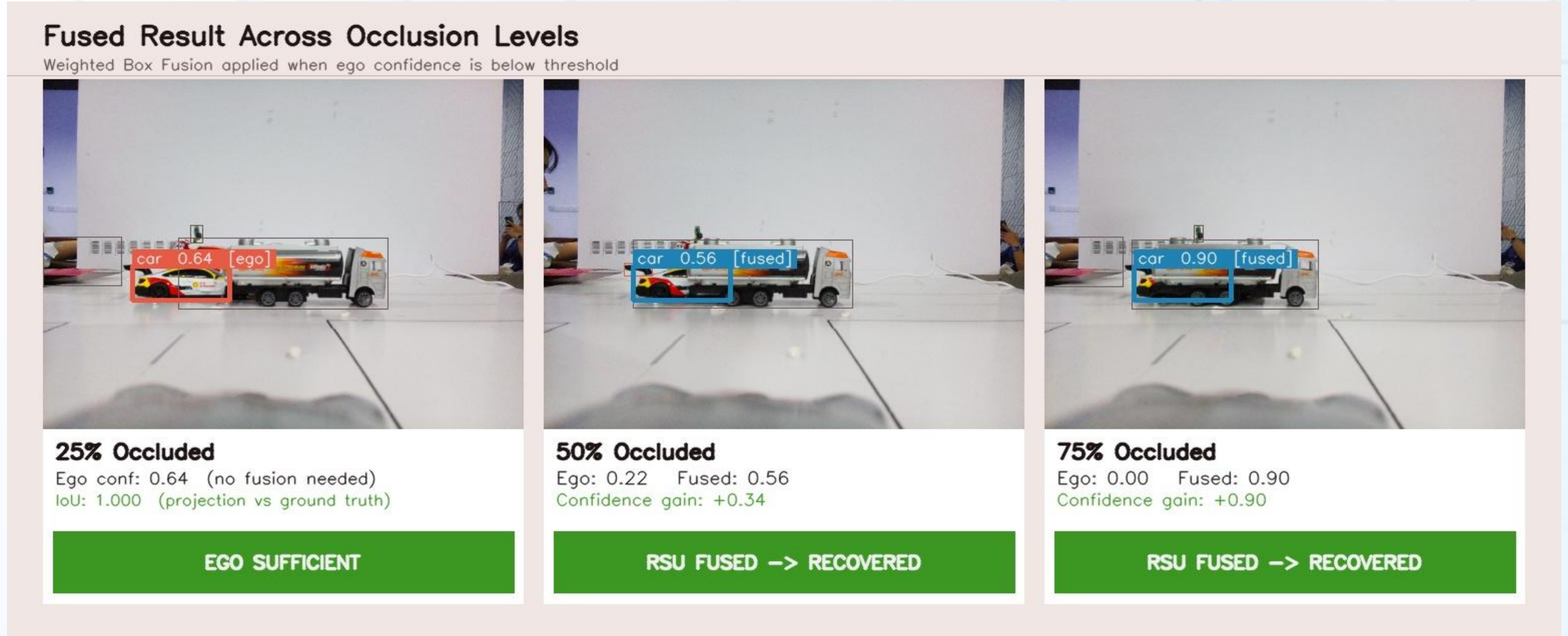

Figure 3: Result of confidence-triggered occlusion recovery

RSU activates when ego confidence drops below the threshold, recovering hidden vehicle detection from **0.00 → 0.88 at 75% occlusion**.

Cooperative Queue Perception

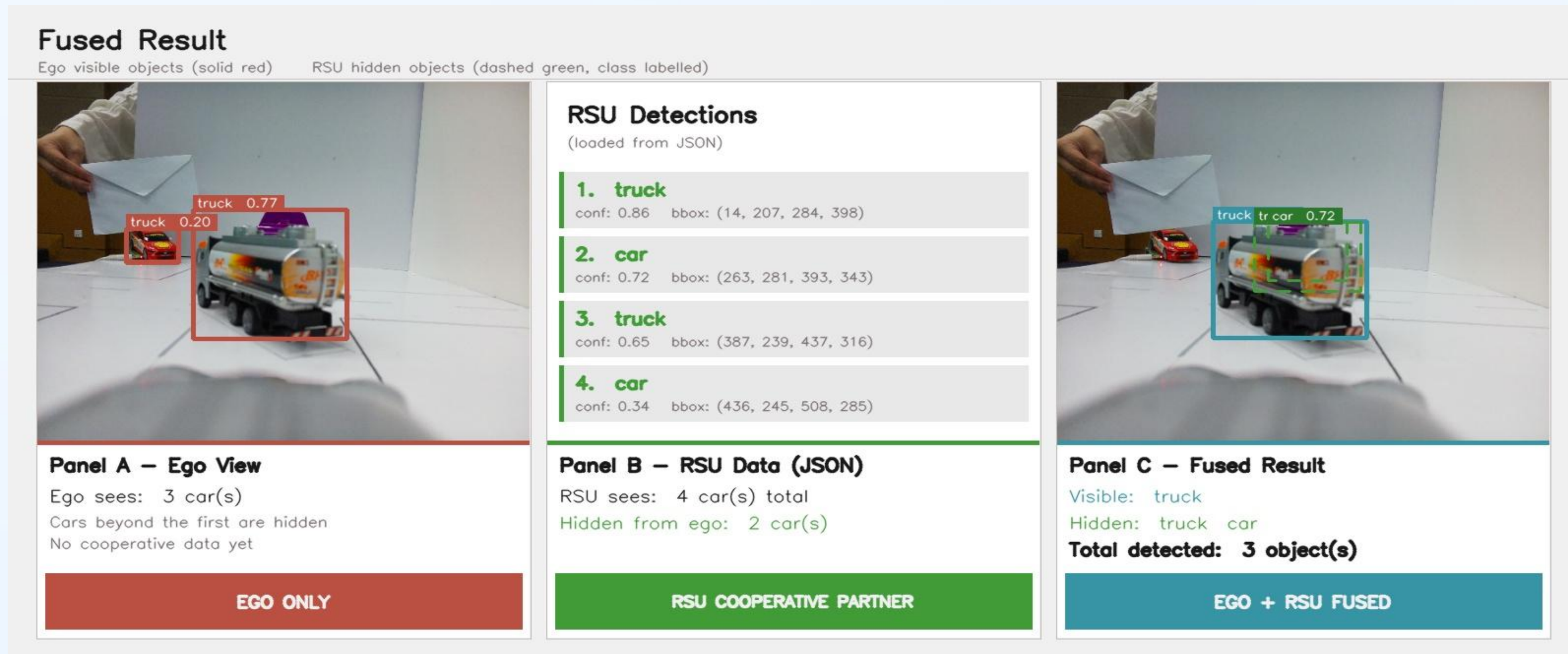

Figure 4: Result of cooperative queue perception

RSU reveals vehicles hidden beyond the ego's line-of-sight, improving queue awareness and driving decision-making.

Communication Efficiency

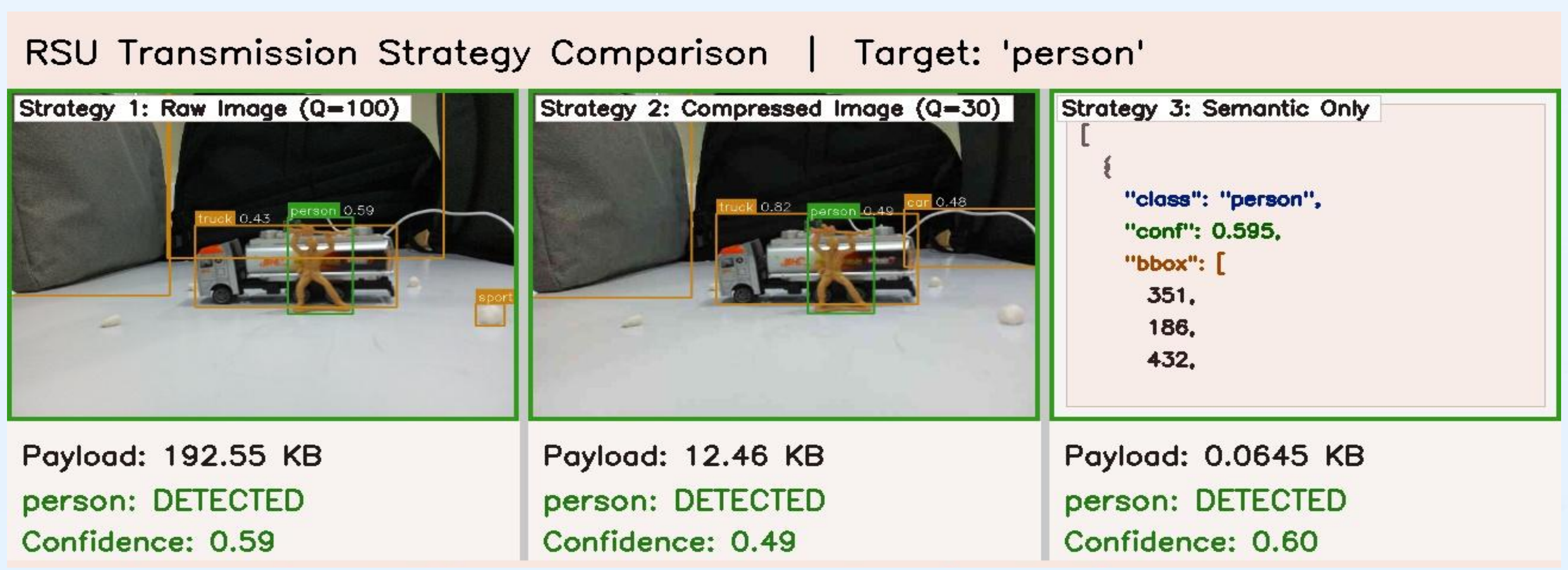

Figure 5: Result of transmission strategy comparison

Semantic-only transmission preserves detection while reducing payload from **192.55 KB → 0.0645 KB (99.9%)**.

06 Conclusion & Next Steps

This project demonstrated that cooperative perception can recover hidden vehicle detections beyond line-of-sight while reducing communication overhead through semantic transmission. Future work includes multi-RSU cooperation, live video streaming and defining real-time delay thresholds.

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